

An introduction to molecular dynamics

From Newton's equations to rare events

Urbain Vaes
urbain.vaes@inria.fr

MATERIALS – Inria Paris & CERMICS – École des Ponts ParisTech

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Outline:

Introduction to molecular dynamics

Time discretization

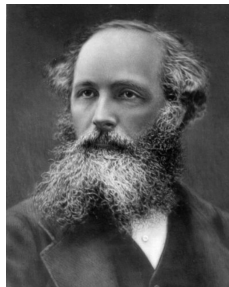
Metastability and rare events

The narrow escape problem

Statistical physics and the atomistic view of matter



Ludwig Boltzmann (1844–1906)



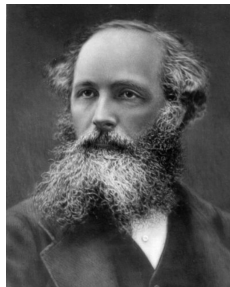
James Clerk Maxwell (1831–1879)

- ▶ At the microscopic scale, matter consists of interacting **atoms**.
- ▶ These interactions give rise to the properties of matter at the macroscopic scale.

Statistical physics and the atomistic view of matter



Ludwig Boltzmann (1844–1906)



James Clerk Maxwell (1831–1879)

- ▶ At the microscopic scale, matter consists of interacting **atoms**.
- ▶ These interactions give rise to the properties of matter at the macroscopic scale.

How do microscopic mechanical laws produce macroscopic behavior? (Hilbert's sixth problem)

Microscopic scale

Newton

$$M\ddot{q} = -\nabla V(q)$$

Trajectories of all particles

$\Downarrow$ many-particle limit

Kinetic scale

Boltzmann

$$\partial_t f + v \cdot \nabla_x f = Q(f, f)$$

One-particle density $f(t, x, v)$

$\Downarrow$ hydrodynamic limit

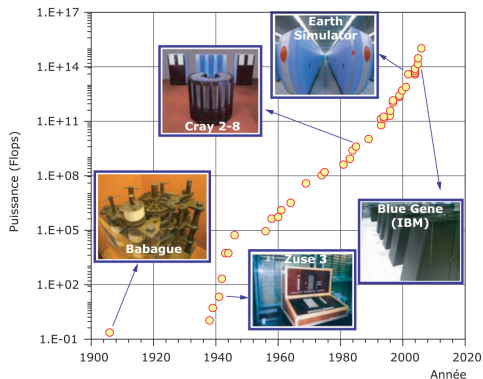
Macroscopic scale

Navier–Stokes

$$\partial_t u + u \cdot \nabla u - \nu \Delta u + \nabla p = 0$$

$\nabla \cdot u = 0$
Fluid velocity and pressure

A second revolution...



Historical computing-power data.

Computing power has increased by a factor of 10^{15} over the past 70 years!

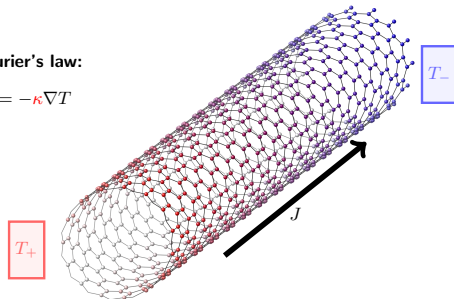
- ▶ Since the 1950s, molecular dynamics has increasingly been used as a **numerical microscope**.
- ▶ Simulation now complements theory and experiment as a **third way of doing science**.

General aims of computational statistical physics

- ▶ Explore matter at the atomic scale and connect microscopic mechanisms to macroscopic properties
- ▶ Predict **static and dynamical properties** and help design new molecules and materials
- ▶ Applications in **chemistry, biology, and materials science**

Fourier's law:

$$J = -\kappa \nabla T$$



“Given the structure and the **laws of interaction** of the particles, what are the **macroscopic properties** of the systems composed of these particles?”

► **Atomistic description:**

- Positions $q = (\mathbf{q}_1, \dots, \mathbf{q}_N)$ and momenta $p = (\mathbf{p}_1, \dots, \mathbf{p}_N)$
- System energy:

$$H(q, p) = V(q) + \frac{1}{2} p^T M^{-1} p$$

- All the physics is encoded in the potential V

► **Macroscopic description at equilibrium:** described by a probability measure μ such as the Boltzmann–Gibbs distribution:

$$\mu(\mathrm{d}q \, \mathrm{d}p) = \frac{1}{Z} \exp(-\beta H(q, p)) \, \mathrm{d}q \, \mathrm{d}p \quad \beta = \frac{1}{k_B T}$$

Equations of motion: fixed energy

For an isolated system, we use Newton's equations:

$$\begin{cases} dq_t = M^{-1} p_t dt, \\ dp_t = -\nabla V(q_t) dt, \end{cases}$$

where M is the mass matrix and $(q_t, p_t) \in \mathbb{R}^{3N} \times \mathbb{R}^{3N}$.

Fundamental property: energy is conserved

$$\frac{d}{dt} H(q_t, p_t) = 0.$$

The equations have a particular symplectic structure:

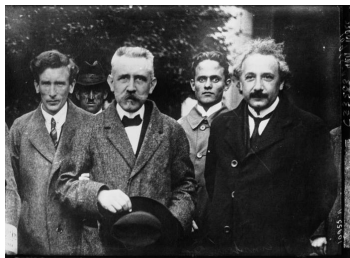
$$\begin{cases} dq_t = \nabla_p H(q_t, p_t) dt, \\ dp_t = -\nabla_q H(q_t, p_t) dt. \end{cases}$$

Langevin dynamics: fixed temperature

For a system coupled to an environment at temperature T , Newton's equations are supplemented with a thermostat:

$$\begin{cases} dq_t = M^{-1} p_t dt, \\ dp_t = -\nabla V(q_t) dt - \gamma M^{-1} p_t dt + \sqrt{2\gamma\beta^{-1}} dW_t. \end{cases}$$

Here $\gamma > 0$ is the friction coefficient and $\beta^{-1} = k_B T$. The **friction** removes energy, while the **random kicks** model thermal agitation.



Source gallica.bnf.fr / Bibliothèque nationale de France

Paul Langevin and Albert Einstein (foreground). Source: BnF/Gallica.

$$\begin{cases} dq_t = M^{-1} p_t dt, \\ dp_t = -\nabla V(q_t) dt - \gamma M^{-1} p_t dt + \sqrt{2\gamma\beta^{-1}} dW_t. \end{cases}$$

Under this dynamics,

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \frac{p_t^T M^{-1} p_t}{2} dt = \frac{3N}{2} k_B T,$$

and, for any integrable observable $\varphi : \mathbb{R}^{3N} \rightarrow \mathbb{R}$,

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \varphi(q_t) dt = \int_{\mathbb{R}^{3N}} \varphi(q) \frac{1}{Z} \exp(-\beta V(q)) dq.$$

The probability measure

$$\frac{1}{Z} \exp(-\beta V(q)) dq, \quad Z = \int_{\mathbb{R}^{3N}} \exp(-\beta V(q)) dq,$$

is the **Boltzmann–Gibbs measure**.

The potential V encodes **all interactions** between atoms

$$V(q) = V_{\text{bond}}(q) + V_{\text{angle}}(q) + V_{\text{dihedral}}(q) + V_{\text{non-bonded}}(q)$$

- ▶ V_{bond} : bond stretching
- ▶ V_{angle} : angle bending
- ▶ V_{dihedral} : torsion angles
- ▶ $V_{\text{non-bonded}}$: non-bonded interactions (van der Waals and electrostatic interactions)

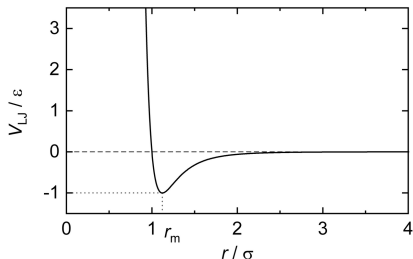
An accurate description of the potential is **essential for reliable predictions!**

Methods for obtaining the potential V :

- ▶ Quantum chemistry calculations
- ▶ Empirical potentials
- ▶ Machine-learned potentials

Example potential: Lennard–Jones

$$V(q) = \sum_{1 \leq i < j \leq N} V_{\text{LJ}}(\|\mathbf{q}_i - \mathbf{q}_j\|)$$
$$V_{\text{LJ}}(r) := 4\varepsilon \left(\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right)$$



Physical scales

- ▶ Distance: from 10^{-10} m (atomic scale) to 10^{-7} m (complex systems)
- ▶ Time from 10^{-15} s (vibrations) to 10^{-3} s (conformational changes)
- ▶ Atomic mass: typically 10^{-27} – 10^{-25} kg
- ▶ Number of particles: 6.02×10^{23} (Avogadro's number)

Simulation scales

- ▶ Time step: $\sim 10^{-15}$ s (to resolve vibrations)
- ▶ Total simulation time: $\sim 10^{-6}$ s
- ▶ Typical system size: 10^3 – 10^6 atoms

Analogy to understand the difference in scales

- ▶ About 8×10^{22} moles of water on Earth
- ▶ One cubic metre of water contains about 3×10^{28} molecules

Molecular simulations (quantum + classical) consume a **large fraction** of supercomputing resources

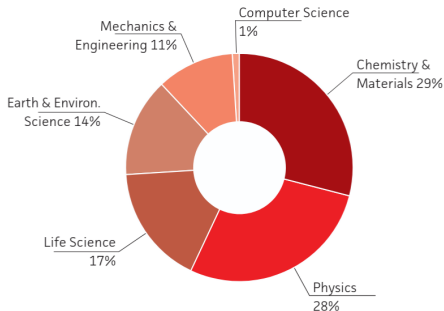


Figure: Usage statistics of the Swiss National Supercomputing Centre (CSCS)

Parallelization strategies

- ▶ **In space**: distribute the computation of forces and energies
- ▶ but **time** is intrinsically sequential $\rightsquigarrow$ limited parallelization

Chemistry

- ▶ Identify reaction mechanisms
- ▶ Design catalysts for applications such as biomass conversion

Biology

- ▶ Characterize protein conformations and transitions
- ▶ Predict ligand–target binding affinities
- ▶ Understand molecular mechanisms in cells

Materials science

- ▶ Study defect propagation and irradiation damage
- ▶ Predict macroscopic properties of new materials

Application I: calculation of static properties of matter (1/2)

Goal: calculate **high-dimensional** averages of the form

$$\mathbf{E}_\mu[\varphi] = \int_{\mathbb{T}^{3N} \times \mathbb{R}^{3N}} \varphi(q, p) \mu(dq dp)$$

- ▶ This problem inspired the development of Markov chain Monte Carlo methods

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VOLUME 21, NUMBER 6

JUNE, 1953

Equation of State Calculations by Fast Computing Machines

NICHOLAS METROPOLIS, ARIANNA W. ROSENBLUTH, MARSHALL N. ROSENBLUTH, AND AUGUSTA H. TELLER,
Los Alamos Scientific Laboratory, Los Alamos, New Mexico

AND

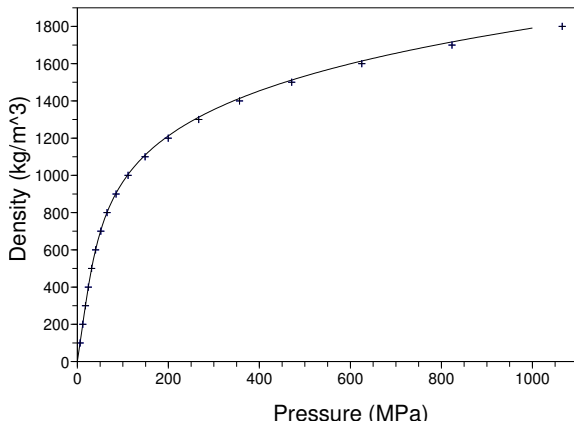
EDWARD TELLER,* *Department of Physics, University of Chicago, Chicago, Illinois*
(Received March 6, 1953)

- ▶ Ensemble averages are approximated by **time averages** along trajectories (ergodicity):

$$\widehat{\varphi}_T = \frac{1}{T} \int_0^T \varphi(q_t, p_t) dt \xrightarrow{T \rightarrow \infty} \mathbf{E}_\mu[\varphi]$$

- ▶ **Mathematical questions:** time discretization, stability, bias, convergence rate, variance reduction.

Application I: calculation of static properties of matter (2/2)



Equation of state for argon at 300 K.

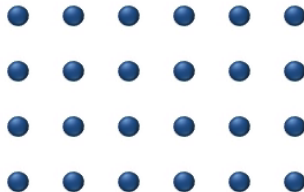
- '+': molecular simulations;
- Solid line: experimental measurements.

Figure: Source: T. Lelièvre, M. Rousset, and G. Stoltz. Imperial College Press, London, 2010

Application II: calculation of dynamical properties of matter

Examples of transport coefficients

- ▶ Mobility of ions in a solution
- ▶ Diffusion coefficients
- ▶ Shear viscosity
- ▶ Thermal conductivity



Langevin dynamics with forcing of amplitude η

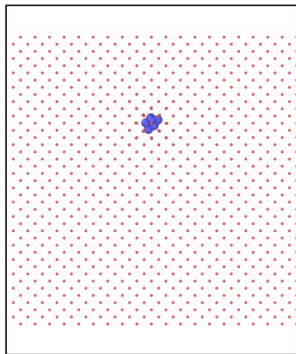
$$dq_t = M^{-1} p_t dt$$

$$dp_t = -\nabla V(q_t) dt + \eta F(q_t) dt - \gamma M^{-1} p_t dt + \sqrt{2\gamma\beta^{-1}} dW_t$$

Transport coefficients are given by the linear response of appropriate observables φ :

$$\alpha = \lim_{\eta \rightarrow 0} \frac{\mathbf{E}_{\mu_\eta}[\varphi] - \mathbf{E}_\mu[\varphi]}{\eta}$$

Unlike static properties, dynamical properties **depend on the precise time evolution**



Helium-induced blistering in tungsten

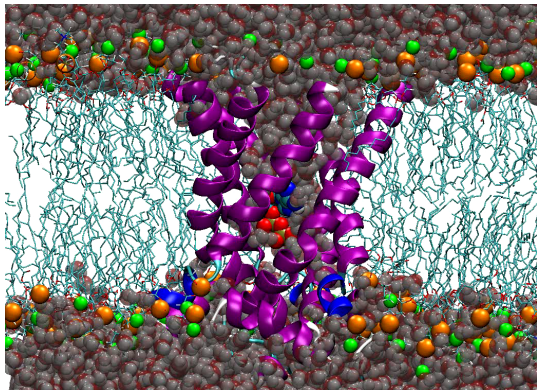
Simulation courtesy of D. Perez (LANL).

A tokamak wall at the atomic scale

Helium irradiation creates nanobubbles beneath the tungsten surface. Their growth can lead to blistering and alter the mechanical properties of the component.

- ▶ Rare nucleation and growth events
- ▶ Accelerated using Parallel Replica dynamics

Application IV: biomolecular conformational change



Mitochondrial adenosine diphosphate (ADP) transporter

Simulation courtesy of C. Chipot (CNRS).

Why enhanced sampling?

Biomolecular systems may contain millions of atoms. Even with an empirical force field, conformational changes occur on time scales inaccessible to a naive trajectory.

- The adaptive biasing force method promotes the slow transition

Application V: protein folding and conformational changes

Studying the various **conformations** and **transitions** between them helps us to

- ▶ explain protein function
- ▶ understand the cause of certain diseases
- ▶ develop new drugs

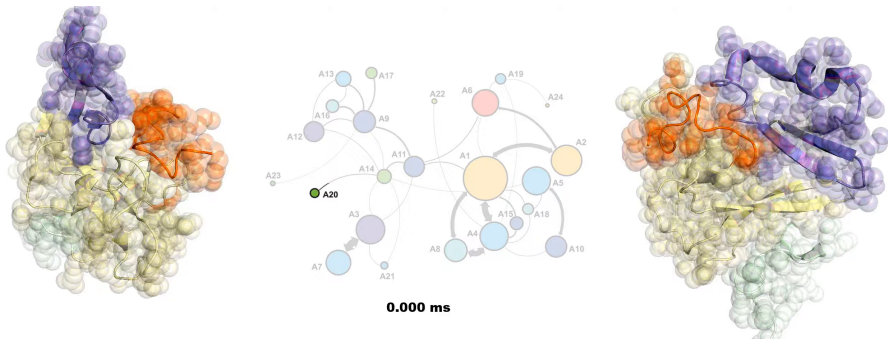


Figure: Source: S. Chen, R. P. Wiewiora, F. Meng, N. Babault, A. Ma, W. Yu, K. Qian, H. Hu, H. Zou, J. Wang, et al. [Elife](#), 2019

Mathematical questions of interest

- ▶ Establish a rigorous link between models at different scales
- ▶ Study and improve algorithms (error analysis, variance reduction, etc.)
- ▶ Develop new algorithms, e.g., for rare events
- ▶ Use data assimilation techniques, e.g., to learn good reaction coordinates

Outline

Introduction to molecular dynamics

Time discretization

Metastability and rare events

The narrow escape problem

To simulate a trajectory, fix $h > 0$ and discretize the equations at times $t_n = nh$, $n \in \mathbb{N}$.

Example 1: Newton's equations

$$\begin{cases} dq_t = M^{-1} p_t dt, \\ dp_t = -\nabla V(q_t) dt \end{cases} \implies \begin{cases} q_{t_{n+1}} - q_{t_n} = \int_{t_n}^{t_{n+1}} M^{-1} p_t dt, \\ p_{t_{n+1}} - p_{t_n} = - \int_{t_n}^{t_{n+1}} \nabla V(q_t) dt. \end{cases}$$

... explicit Euler discretization ...

$$\begin{cases} q^{n+1} - q^n = M^{-1} p^n h, \\ p^{n+1} - p^n = -\nabla V(q^n) h. \end{cases}$$

Here $(q^0, p^0) = (q_0, p_0)$, and we hope that (q^n, p^n) is a good approximation of (q_{t_n}, p_{t_n}) .

Example 2: Langevin equations

$$\begin{cases} dq_t = M^{-1} p_t dt, \\ dp_t = -\nabla V(q_t) dt + \left(-\gamma M^{-1} p_t dt + \sqrt{2\gamma\beta^{-1}} dW_t \right). \end{cases}$$

... Euler-Maruyama discretization ...

$$\begin{cases} q^{n+1} - q^n = M^{-1} p^n h, \\ p^{n+1} - p^n = -\nabla V(q^n) h + \left(-\gamma M^{-1} p^n h + \sqrt{2\gamma\beta^{-1}} h G^n \right). \end{cases}$$

Here $G^n = (G_1^n, \dots, G_{3N}^n)$ and G_i^n are independent standard Gaussian random variables.

- For intuition, one may think of $G_i^n = \pm 1$ with probability $1/2$.
- Implementation requires generating independent random numbers

The Verlet scheme¹

For Newtonian dynamics, a standard alternative to explicit Euler is the **Verlet scheme**:

$$\begin{cases} p^{n+\frac{1}{2}} - p^n = -\frac{1}{2} \nabla V(q^n) h, \\ q^{n+1} - q^n = M^{-1} p^{n+\frac{1}{2}} h, \\ p^{n+1} - p^{n+\frac{1}{2}} = -\frac{1}{2} \nabla V(q^{n+1}) h. \end{cases}$$

- ▶ Second-order accurate
- ▶ Time-reversible
- ▶ Symplectic

Why does Verlet keep the energy nearly constant over much longer times?

→ Backward error analysis

¹E. Hairer, C. Lubich, and G. Wanner. [Acta Numer.](#), 2003.

The Verlet scheme: a modified Hamiltonian

$$\begin{cases} p^{n+\frac{1}{2}} - p^n = -\frac{1}{2} \nabla V(q^n) h, \\ q^{n+1} - q^n = M^{-1} p^{n+\frac{1}{2}} h, \\ p^{n+1} - p^{n+\frac{1}{2}} = -\frac{1}{2} \nabla V(q^{n+1}) h. \end{cases}$$

Backward error analysis

For every $k \geq 1$, there is a modified Hamiltonian $\tilde{H}$ such that, for $t \in [t_n, t_{n+1}]$, the solution of

$$\begin{cases} d\tilde{q}_t = \nabla_p \tilde{H}(\tilde{q}_t, \tilde{p}_t) dt, \\ d\tilde{p}_t = -\nabla_q \tilde{H}(\tilde{q}_t, \tilde{p}_t) dt, \end{cases} \quad (\tilde{q}_{t_n}, \tilde{p}_{t_n}) = (q^n, p^n),$$

satisfies

$$\|q^{n+1} - \tilde{q}_{t_{n+1}}\| + \|p^{n+1} - \tilde{p}_{t_{n+1}}\| \leq C_k h^{2k+2}.$$

Moreover,

$$\tilde{H} = H + h^2 H_2 + \cdots + h^{2k} H_{2k}.$$

The Verlet scheme: consequences

Since the modified flow conserves $\tilde{H}$ exactly, the previous estimate gives

$$\begin{aligned}\tilde{H}(q^{n+1}, p^{n+1}) &= \tilde{H}(\tilde{q}_{t_{n+1}}, \tilde{p}_{t_{n+1}}) + \mathcal{O}(h^{2k+2}) \\ &= \tilde{H}(q^n, p^n) + \mathcal{O}(h^{2k+2}).\end{aligned}$$

Summing this one-step error over n steps gives

$$\tilde{H}(q^n, p^n) = \tilde{H}(q^0, p^0) + \mathcal{O}(nh^{2k+2}).$$

Since $\tilde{H} = H + \mathcal{O}(h^2)$,

$$H(q^n, p^n) = H(q^0, p^0) + \mathcal{O}(h^2) + \mathcal{O}(nh^{2k+2}).$$

Long-time behavior

The physical energy remains within $\mathcal{O}(h^2)$ over times

$$nh \leqslant C_k h^{1-2k}.$$

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Metastability and “the time scale problem”

For many systems of interest, there are two **widely separated** time scales:

- ▶ **Fast time scale** $\sim 10^{-15}$ s: vibrations of atoms
- ▶ **Slow time scale**: conformational changes, chemical reactions $> 10^{-6}$ s

This is a manifestation of **metastability** of the dynamics. **A metastable state** is a region of configuration space where the system remains for long periods before transitioning

- ▶ Fast relaxation to local equilibrium within a state
- ▶ Slow transitions between states (rare events)

$\leadsto$ **computational challenge** for sampling!

Two types of metastability:

- ▶ **Energetic origin**: states are separated by high-energy barriers
- ▶ **Entropic origin**: states are separated by narrow passages in configuration space

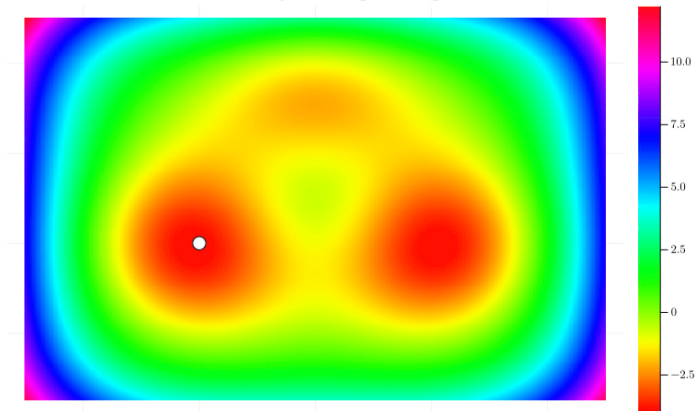
Metastability of energetic origin

Consider Langevin dynamics in two dimensions with a **multi-well potential** V :

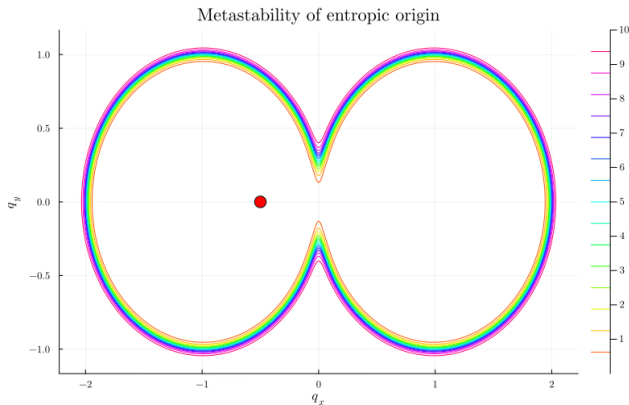
$$dq_t = p_t dt$$

$$dp_t = -\nabla V(q_t) dt - \gamma p_t dt + \sqrt{2\gamma\beta^{-1}} dW_t$$

Metastability of energetic origin



Metastability of entropic origin



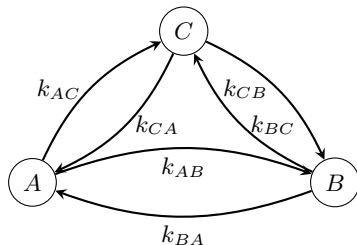
Mathematical questions of interest:

- ▶ What is the expected time to leave a metastable state?
- ▶ What is the distribution of the exit time?
- ▶ What is the distribution of the exit point?

Simplified Markov state model: the kinetic Monte Carlo approach

When the system is metastable, the dynamics can be approximated by a **Markov jump process** between metastable states. **Idea:**

- ▶ Each state = **metastable region** in configuration space
- ▶ Transitions = **rare events** with rates k_{ij}



Mathematical questions:

- ▶ How to parameterize the Markov model from the original dynamics?
- ▶ In what setting is this a good approximation?
- ▶ Are there more accurate methods to approximate the state-to-state dynamics?

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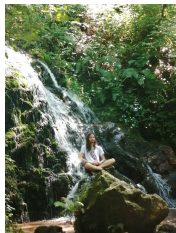
The narrow escape problem

Scientific collaborators

Louis Carillo



Lois Delande



Tony Lelièvre



Thomas Normand



Gabriel Stoltz



The narrow escape problem

- ▶ Domain Ω_ε with small doors:

$$\Omega_\varepsilon = \Omega \setminus \bigcup_{n=1}^N \overline{B}(x_n, r_n^\varepsilon)$$

- ▶ Brownian particle:

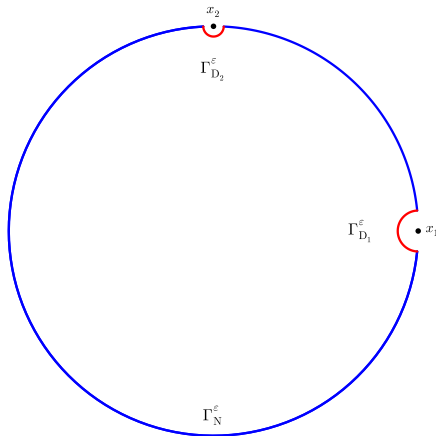
$$dX_t = \sqrt{2} dW_t + \text{reflection}$$

- ▶ Reflecting boundary Γ_N^ε
- ▶ N exit doors $\Gamma_{D_1}^\varepsilon, \dots, \Gamma_{D_N}^\varepsilon$

Let $\tau_\varepsilon := \inf\{t \geq 0, X_t \in \Gamma_D^\varepsilon\}$ with

$$\Gamma_D^\varepsilon = \bigcup_{n=1}^N \Gamma_{D_n}^\varepsilon$$

Objective: Characterize the **first-exit event** $(\tau_\varepsilon, X_{\tau_\varepsilon})$ in the limit $\varepsilon \rightarrow 0$



Motivations from biology and chemistry:

- ▶ Escape of ions through small openings in cell membranes
- ▶ Early stages of viral infection
- ▶ Binding of diffusing molecules to active sites

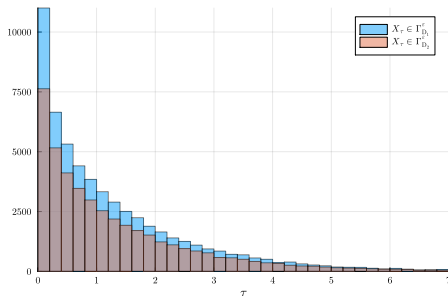
Vast existing literature, but few rigorous results until recently:

- ▶ D. Holcman and Z. Schuss. *J. Stat. Phys.*, 2004
- ▶ O. Bénichou and R. Voituriez. *Phys. Rev. Letters*, 2008
- ▶ H. Ammari, K. Kalimeris, H. Kang, and H. Lee. *J. Math. Pures Appl.* (9), 2012
- ▶ D. Holcman and Z. Schuss. *SIAM Rev.*, 2014
- ▶ X. Chen and A. Friedman. *SIAM J. Math. Anal.*, 2011

Preliminary numerical experiments (1/2)

We simulate $M = 10^5$ Brownian paths from $X_0 \sim \mathcal{U}(\Omega_\varepsilon)$ until they escape:

- ▶ Exit-door radii: $r_1^\varepsilon = 0.1$ and $r_2^\varepsilon = 0.05$
- ▶ We record $(\tau_\varepsilon, \text{index}(X_{\tau_\varepsilon}))$ for each exit event



Observations:

- ▶ Escape time τ_ε appears to follow an **exponential distribution**
- ▶ The index of the exit door appears to be **independent of τ_ε**

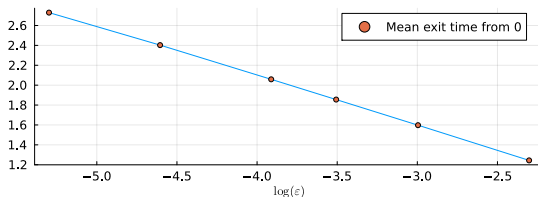
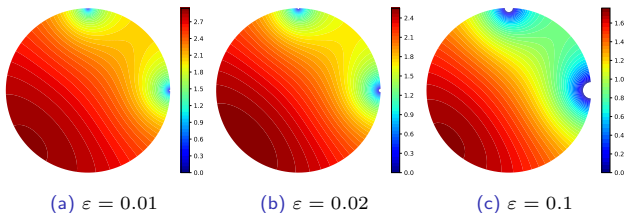
Preliminary numerical experiments (2/2)

Link with a partial differential equation

Let $T_\varepsilon : \Omega_\varepsilon \rightarrow \mathbf{R}$ be defined by $T_\varepsilon(x) := \mathbf{E}_x[\tau_\varepsilon]$. Then T_ε satisfies

$$-\Delta T_\varepsilon = 1 \quad \text{in } \Omega_\varepsilon, \quad \partial_n T_\varepsilon = 0 \quad \text{on } \Gamma_N^\varepsilon \quad T_\varepsilon = 0 \quad \text{on } \Gamma_D^\varepsilon$$

Finite element approximation of the mean exit time from x (with $r_1^\varepsilon = 2r_2^\varepsilon = \varepsilon$)



Numerical experiments suggest that the exit event $(\tau_\varepsilon, X_{\tau_\varepsilon})$ has the following properties:

- ▶ The exit time τ_ε is approximately **exponentially distributed**
- ▶ The exit point X_{τ_ε} is **approximately independent** of the exit time τ_ε
- ▶ In two dimensions, the mean exit time diverges as $\log(\varepsilon^{-1})$ as $\varepsilon \rightarrow 0$

Mathematical analysis:

- ▶ Can we rigorously prove these properties?
- ▶ What is the distribution of the exit point?
- ▶ What is the impact of the starting point X_0 on the exit event?
- ▶ What changes in higher dimensions?

The quasi-stationary distribution approach

- ▶ **Goal:** Rewrite the narrow escape problem as a spectral problem
- ▶ **Motivation:** For $\varepsilon \ll 1$, the particle reaches local equilibrium before leaving

Definition: quasi-stationary distribution^{1,2}

The QSD ν_ε is a probability measure on Ω_ε such that

$$\forall t \geq 0, \quad X_0 \sim \nu_\varepsilon \quad \Rightarrow \quad \mathcal{L}aw(X_t \mid \tau_\varepsilon > t) = \nu_\varepsilon$$

Property (Yaglom limit): For any initial point $x \in \Omega_\varepsilon$ and any measurable set A ,

$$\lim_{t \rightarrow \infty} \mathbf{P}_x [X_t \in A \mid \tau_\varepsilon > t] = \nu_\varepsilon(A)$$

¹S. Méléard and D. Villemonais. *Probab. Surv.*, 2012.

²T. Lelièvre and G. Stoltz. *Acta Numer.*, 2016, Section 6.3.1.

Illustration of the Yaglom limit

Fokker–Planck equation:

$$\begin{cases} \partial_t \rho_\varepsilon = \Delta \rho_\varepsilon & \text{in } \Omega_\varepsilon \\ \partial_n \rho_\varepsilon = 0 & \text{on } \Gamma_N^\varepsilon \\ \rho_\varepsilon = 0 & \text{on } \Gamma_D^\varepsilon \end{cases}$$

gives evolution of $\mathcal{L}aw(X_t \mid \tau_\varepsilon > t)$

On the right, $X_0 \sim \mathcal{N}(0, 0.04I_2)$

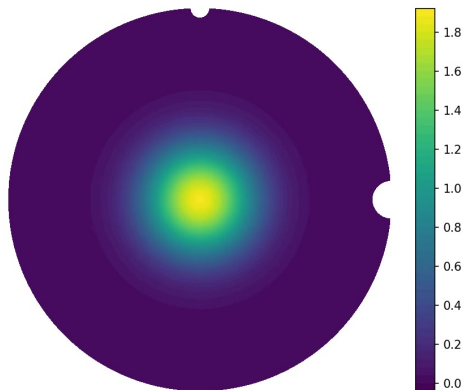


Figure: Evolution of $\mathcal{L}aw(X_t \mid \tau_\varepsilon > t)$

Assume that $X_0 \sim \nu_\varepsilon$. Then:

- ▶ The exit time τ is **exponentially distributed** $\sim \text{Exp}(\lambda_\varepsilon)$

$$\begin{aligned}\mathbf{P}_{\nu_\varepsilon}[\tau_\varepsilon \geq s + t] &= \mathbf{P}_{\nu_\varepsilon}[\tau_\varepsilon \geq s + t \mid \tau_\varepsilon \geq s] \mathbf{P}_{\nu_\varepsilon}[\tau_\varepsilon \geq s] \\ &= \mathbf{P}_{\nu_\varepsilon}[\tau_\varepsilon \geq t] \mathbf{P}_{\nu_\varepsilon}[\tau_\varepsilon \geq s].\end{aligned}$$

- ▶ The exit point X_{τ_ε} is **independent** of the exit time τ_ε

$$\begin{aligned}\mathbf{P}_{\nu_\varepsilon}[X_{\tau_\varepsilon} \in A, \tau_\varepsilon \geq t] &= \mathbf{P}_{\nu_\varepsilon}[X_{\tau_\varepsilon} \in A \mid \tau_\varepsilon \geq t] \mathbf{P}_{\nu_\varepsilon}[\tau_\varepsilon \geq t] \\ &= \mathbf{P}_{\nu_\varepsilon}[X_{\tau_\varepsilon} \in A] \mathbf{P}_{\nu_\varepsilon}[\tau_\varepsilon \geq t]\end{aligned}$$

Goal: Study λ_ε and $\mathbf{P}_{\nu_\varepsilon}[X_{\tau_\varepsilon} \in \Gamma_{D_i}^\varepsilon]$ for $i \in \{1, \dots, N\}$ in the limit $\varepsilon \rightarrow 0$

Explicit formulas for the mean exit time and exit-point distribution

Spectral characterization of the QSD

Consider the first eigenpair $(u_\varepsilon, \lambda_\varepsilon)$ of the eigenvalue problem

$$\begin{cases} -\Delta u_\varepsilon = \lambda_\varepsilon u_\varepsilon & \text{in } \Omega_\varepsilon \\ \partial_n u_\varepsilon = 0 & \text{on } \Gamma_N^\varepsilon \\ u_\varepsilon = 0 & \text{on } \Gamma_D^\varepsilon \end{cases}$$

Then the QSD is given by

$$\nu_\varepsilon(dx) = \frac{u_\varepsilon(x) dx}{\int_{\Omega_\varepsilon} u_\varepsilon(y) dy}$$

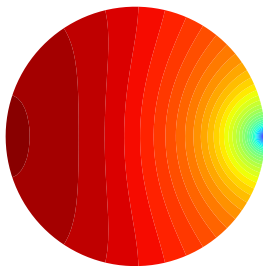
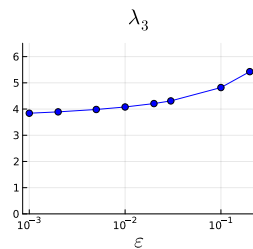
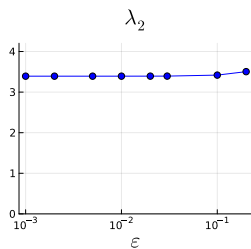
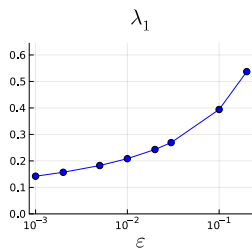
(The first eigenfunction can be chosen strictly positive, and $\lambda_\varepsilon > 0$.)

- ▶ The mean exit time satisfies $\mathbf{E}_{\nu_\varepsilon}[\tau_\varepsilon] = \frac{1}{\lambda_\varepsilon}$
- ▶ The distribution of the exit point satisfies

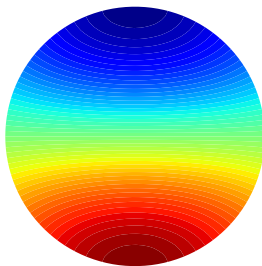
$$\mathbf{P}_{\nu_\varepsilon}[X_{\tau_\varepsilon} \in \Gamma_{D_i}^\varepsilon] = \frac{\int_{\Gamma_{D_i}^\varepsilon} \partial_n u_\varepsilon d\sigma}{\int_{\Gamma_D^\varepsilon} \partial_n u_\varepsilon d\sigma}$$

→ The QSD provides information about both the exit time **and the exit point**

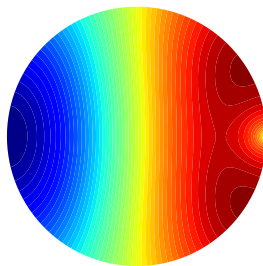
Numerical illustration of the eigenfunctions



(a) First eigenfunction



(b) Second eigenfunction



(c) Third eigenfunction

Theorem

Under appropriate assumptions, as $\varepsilon \rightarrow 0$,

$$\mathbf{E}[\tau_\varepsilon] \sim C_{\Omega,d} \frac{1}{\overline{K}_\varepsilon} \quad \text{and} \quad \forall i \in \{1, \dots, N\}, \quad \mathbf{P}[X_{\tau_\varepsilon} \in \Gamma_{D_i}^\varepsilon] \sim \frac{K_\varepsilon^i}{\overline{K}_\varepsilon}$$

where

$$K_\varepsilon^i := \begin{cases} -\frac{1}{\log(r_i^\varepsilon)} & \text{if } d = 2 \\ (r_i^\varepsilon)^{d-2} & \text{if } d \geq 3 \end{cases} \quad \overline{K}_\varepsilon := K_\varepsilon^1 + \dots + K_\varepsilon^N$$

Numerical simulations are extremely useful for illustrating the results and gaining intuition:

- ▶ Finite element simulation in 2D and 3D
- ▶ Monte Carlo simulation, accelerated by a [walk-on-spheres](#) algorithm
- ▶ Fleming–Viot particle system to approximate the quasi-stationary distribution

Molecular-scale simulation raises many questions in **modeling** and **algorithm design** to which mathematics can make essential contributions.

Topics not covered today include:

- ▶ Connections with Monte Carlo methods in statistics
- ▶ Electronic-structure calculations used to parameterize force fields
- ▶ The growing role of **machine learning**: dimension reduction, force fields, AlphaFold, and generative models

A strength of applied mathematics: interdisciplinarity

Applied mathematics draws strength from connecting other sciences (biology, chemistry, mechanics, materials science, etc.) and mathematical fields (analysis, PDEs, probability, numerical analysis, etc.).

Thank you for your attention!



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